

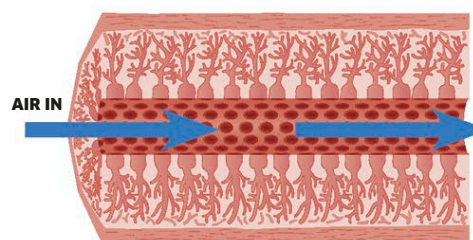
Getting air

Birds need plenty of energy to fuel active lives. Their beautifully efficient system for getting oxygen to cells is unique—no other animal has one quite like it.

The key to a bird's breathing system lies in big, air-filled sacs that pack the body. They help supply air to the lungs, which—unlike human lungs—are small and rigid. Breathing makes the sacs inflate and deflate like balloons, sweeping fresh air through the lungs. Oxygen continually seeps into the blood and circulates to the cells, so they can release energy in respiration.

Abdominal air sacs
The biggest pair of air sacs are in the abdomen (belly) of the body.

Parabronchi (air-filled tubes) bring air to the air vessels.



Microscopic tubes called capillaries carry blood.

Gaseous exchange
Each lung is filled with tiny air-filled vessels intermingled with microscopic blood vessels, helping to bring oxygen as close to the blood as possible.

Breathing at high altitudes

Traveling at high altitudes poses a special problem for some high-flying birds, as the air gets so thin that there is little oxygen. The migration route of bar-headed geese takes them over the Himalayas—the highest any known bird has flown. To cope with this, they have bigger lungs than other waterfowl and can breathe more deeply, while the pigment in their blood (hemoglobin) traps oxygen in the thin air especially well.

